$\boxed{a(\text{graph of } f)=0}$

0.1 rectangle: any set congruent to the set $\{(x,y) \mid 0\le x\le h,\ 0\le y\le k\}$, where $h\geq 0$, $k\geq 0$.

0.2 step region: the union of a finite collection of adjacent rectangles with their bases resting on the x-axis.

0.3 ① $\forall$ step regions S $\Rightarrow \exists$ naturally step function $s\geq 0$, s.t., $a(S)=\int_a^b s(x)dx$

② $\forall$ step function $s\geq 0$, $\exists$ naturally step region S, s.t., $a(S)=\int_a^b s(x)dx$

0.4 set Q is enclosed by some step regions.

$\Rightarrow$ sup, inf $\exists$, and sup $\le$ inf

here $\sup := \sup\{a(S) \mid \text{step region } S\subseteq Q\}$

$\inf := \inf\{a(T) \mid \text{step region } T\supseteq Q\}$

proof: $\because$ Q enclosed by some step regions,

$\therefore \{a(S) \mid S\subseteq Q\}$, $\{a(T) \mid T\supseteq Q\}$ nonempty.

$\because a(S)\le a(T)$, $\forall S\subseteq Q\subseteq T$

$\therefore$ sup, inf $\exists$, and sup $\le$ inf.

0.5 f is defined and bounded on $[a,b]$

$\Rightarrow$ sup, inf $\exists$ and sup $\le$ inf.

here $\sup := \sup\{\int_a^b s(x)dx \mid \text{step function } s\le f\}$

$\inf := \inf\{\int_a^b t(x)dx \mid \text{step function } t\geq f\}$

proof: $\because f$ bounded

$\therefore \{\int s \mid s\le f\}$, $\{\int t \mid t\geq f\}$ nonempty

$\because \int s\le\int t$ if $s\le f\le t$ $\therefore$ sup, inf $\exists$

and sup $\le$ inf

1. $\exists c$, unique, s.t., $a(S)\le c\le a(T)$

$\forall$ step regions S, T, $S\subseteq Q\subseteq T$

$\Leftrightarrow \sup\{a(S) \mid \text{step region } S\subseteq Q\}$

$=\inf\{a(T) \mid \text{step region } T\supseteq Q\}$

(here Q is a set that can be enclosed by some step regions).

proof: 1) $\Rightarrow$

① $\because$ Q enclosed by some step regions

$\therefore$ sup, inf $\exists$, and sup $\le$ inf (from 0.4)

② $\forall c'\in[\sup,\inf]$,

$a(S)\le c'\le a(T)$, $\forall S,T$, $S\subseteq Q\subseteq T$

$\because c'$ must be unique $\therefore$ sup $=$ inf

2) $\Leftarrow$

① $\because$ sup $=$ inf, $\therefore \exists c=\sup=\inf$,

$\forall$ step region $S\subseteq Q$, $a(S)\le c$

$\forall$ step region $T\supseteq Q$, $a(T)\geq c$

② if $\exists c'$, s.t., $a(S)\le c'\le a(T)$, $\forall S\subseteq Q\subseteq T$

then, $\sup\le c'\le\inf$

$\because$ sup $=$ inf $\therefore c'=c$, $\therefore c$ unique

2. axiom

$\exists c$, unique, s.t., $a(S)\le c\le a(T)$

$\forall$ step regions S, T, satisfying $S\subseteq Q\subseteq T$.

$\Rightarrow Q$ measurable $\Rightarrow a(Q)=c$ (monotone property)

(here Q a set that can be enclosed by some S, T, i.e., $\exists$ step regions S, T, $S\subseteq Q\subseteq T$).

2.3 sup $=$ inf, i.e., $\sup\{a(S) \mid \text{step region } S\subseteq Q\}$
$=\inf\{a(T) \mid \text{step region } T\supseteq Q\}$

$\Rightarrow Q$ measurable (use 1)

$\Rightarrow a(Q)=\sup=\inf$ (use monotone property)

(here set Q is enclosed by some step regions)

3. $\exists I$, unique, s.t.,
$\int_a^b s(x)dx \le I \le \int_a^b t(x)dx$
$\forall$ step functions s, t, satisfying
$s(x) \le f(x) \le t(x)$

$\Leftrightarrow$ sup = inf, i.e.,
$\sup\{\int_a^b s(x)dx \mid \text{step function } s \le f\}$
$= \inf\{\int_a^b t(x)dx \mid \text{step function } t \ge f\}$
(here f defined and bounded on $[a,b]$)

proof: $\Rightarrow$

① $\because f$ bounded on $[a,b]$
$\therefore$ sup, inf $\exists$, and sup $\le$ inf (from 0.5)

② $\forall c' \in [\sup, \inf]$,
$\int s \le c' \le \int t$, $\forall s \le f \le t$
$\because c'$ unique $\therefore$ sup = inf

$\Leftarrow$

① $\because$ sup = inf $\therefore \exists I = \sup = \inf$
$\int s \le I \le \int t$, $\forall s, t$, $s \le f \le t$

② if $\exists I'$, s.t., $\int s \le I' \le \int t$, $\forall s \le f \le t$
$\Rightarrow I' \in [\sup, \inf]$

$\because$ sup = inf $\therefore I' = I$ $\therefore I$ unique

4. definition
$\exists I$, unique, s.t., $\int_a^b s(x)dx \le I \le \int_a^b t(x)dx$
$\forall$ step functions s, t, satisfying $s(x) \le f(x) \le t(x)$
$\Leftrightarrow f$ integrable (definition)
$\Rightarrow \int_a^b f(x)dx = I$ (use comparison theorem)
(here f defined and bounded on $[a,b]$)

4.3 sup = inf, i.e.,
$\sup\{\int_a^b s(x)dx \mid \text{step function } s \le f\}$
$= \inf\{\int_a^b t(x)dx \mid \text{step function } t \ge f\}$
$\Leftrightarrow f$ integrable
$\Rightarrow \int f = \sup = \inf$
(here f defined and bounded on $[a,b]$)

5. step function $s \ge 0$, defined on $[a,b]$,
Q ordinate set of s
$\Rightarrow Q$ measurable, $a(Q) = \int_a^b s(x)dx$

6 $f \ge 0$, integrable on $[a,b]$,
Q ordinate set of f over $[a,b]$
(here $Q = \{(x,y) \mid a \le x \le b, 0 \le y \le f(x)\}$)
$\Rightarrow$ ① $\int f = \sup\{a(S) \mid S \subseteq Q, S \text{ step region}\}$
$= \inf\{a(T) \mid T \supseteq Q, T \text{ step region}\}$
② Q measurable, $a(Q) = \int_a^b f(x)dx$

proof: ① f integrable on $[a,b]$
$\Rightarrow f$ defined and bounded on $[a,b]$
$f \ge 0 \Rightarrow Q \,\exists$
$\therefore Q$ is bounded by some step regions.

② $\forall$ step region $S \subseteq Q$, $\exists$ step function $0 \le s \le f$,
s.t., $a(S) = \int s$ (from 0.3)
$\therefore \{a(S) \mid S \subseteq Q\} \subseteq \{\int s \mid 0 \le s \le f\}$
$\forall$ step function $0 \le s \le f$, $\exists$ step region $S \subseteq Q$,
s.t., $a(S) = \int s$ (from 0.3)
$\therefore \{\int s \mid 0 \le s \le f\} \subseteq \{a(S) \mid S \subseteq Q\}$
$\therefore \{\int s \mid 0 \le s \le f\} = \{a(S) \mid S \subseteq Q\}$

② Q' measurable, $a(Q') = \int f$

③ $a(\text{graph of } f) = 0$

proof: $\because \int f = \sup\{a(S) \mid \text{step region } S \subseteq Q\}$

$= \inf\{a(T) \mid \text{step region } T \supseteq Q\}$

(from 6)

$\therefore \int f = \sup\{a(S') \mid S' \subseteq Q'\} = \inf\{a(T') \mid T' \supseteq Q'\}$

(from 7)

$\therefore$ Q' measurable, $a(Q') = \int f$ (from 2, 3)

$\therefore a(\text{graph of } f) = a(Q) - a(Q') = 0$

(from difference property of area axiom)

9. $f \geq 0$, integrable on $[a,b]$, Q', Q from 8

$\Rightarrow$ ① Q' measurable, $a(Q') = \int f$. ② $a(\text{graph of } f) = 0$

proof: let $g(x) = f(x) + 1$ $\therefore g(x) > 0$, g integrable on $[a,b]$.

(basic property of integral)

$\therefore a(\text{graph of } g) = 0$ (from 8)

$\therefore a(\text{graph of } f) = 0$

(invariance under congruence of area axiom)

$\therefore$ Q' measurable, $a(Q') = a(Q) - a(\text{graph of } f)$

$= \int f$ ▨.

10. f integrable on $[a,b] \Rightarrow a(\text{graph of } f) = 0$

proof: ① f integrable $\Rightarrow$ f bounded, let $m \leq f(x) \leq M$

② let $g(x) = f(x) - m$ $\therefore g \geq 0$, g integrable on $[a,b]$

$\therefore a(\text{graph of } g) = 0$ $\therefore a(\text{graph of } f) = 0$ ▨

$\forall \varepsilon > 0, \exists S \subseteq Q, a(S) > \sup - \varepsilon$.

$\therefore \exists S' \subseteq Q', a(S') > \sup - 2\varepsilon$.

$\therefore \sup = \sup'$.

detail: let S: height s_i if $x \in (x_{i-1}, x_i)$

S': height $s_i' = \max(s_i - \delta, 0)$, if $x \in (x_{i-1}, x_i)$

$i = 1, \cdots, n$.

$P = \{x_0, x_1, \cdots, x_n\}$ be partition of $[a,b]$

$\because a(S) - a(S') \leq \delta(b-a)$

$\therefore$ choose $\delta < \frac{\varepsilon}{b-a}$, we have $a(S) - a(S') < \varepsilon$

② $\because Q' \subseteq Q$, $\therefore \{a(T) \mid T \supseteq Q\} \subseteq \{a(T') \mid T' \supseteq Q'\}$

$\forall T' \supseteq Q'$, let T': height t_i if $x \in (x_{i-1}, x_i)$

$\because T' \supseteq Q' = \{(x,y) \mid a \leq x \leq b, 0 \leq y < f(x)\}$

$\therefore t_i \geq f(x)$, if $x \in (x_{i-1}, x_i)$

$\therefore T' \supseteq Q$ $\therefore \{a(T) \mid T \supseteq Q\} \supseteq \{a(T') \mid T' \supseteq Q'\}$

$\therefore \{a(T) \mid T \supseteq Q\} = \{a(T') \mid T' \supseteq Q'\}$

8. $f > 0$, integrable on $[a,b]$, ▨

$Q' = Q - \text{graph of } f$,

Q ordinate set of f over $[a,b]$.

here $Q' = \{(x,y) \mid a \leq x \leq b, 0 \leq y < f(x)\}$

$\Rightarrow$ ① $\int f = \sup\{a(S') \mid \text{step region } S' \subseteq Q'\}$

$= \inf\{a(T') \mid \text{step region } T' \supseteq Q'\}$

③ similarly, $\{\int t \mid t \geq f\} = \{a(T) \mid T \supseteq Q\}$

④ $\because f$ integrable $\Rightarrow \int f = \sup\{\int s \mid 0 \leq s \leq f\}$

$= \inf\{\int t \mid t \geq f\}$

(from 4, 3)

$\therefore \int f = \sup\{a(S) \mid \text{step region } S \subseteq Q\}$

$= \inf\{a(T) \mid \text{step region } T \supseteq Q\}$

$\therefore$ Q measurable, $a(Q) = \int f$ (from 2, 3)

▨

7. $f > 0$, integrable on $[a,b]$,

$Q' = Q - \text{graph of } f$,

Q is ordinate set of f over $[a,b]$

$\Rightarrow$ ① $\sup\{a(S) \mid \text{step region } S \subseteq Q\}$

$= \sup\{a(S') \mid \text{step region } S' \subseteq Q'\}$

② $\{a(T) \mid \text{step region } T \supseteq Q\}$

$= \{a(T') \mid \text{step region } T' \supseteq Q'\}$

proof: ① $\because Q' \subseteq Q$

$\therefore \sup = \sup\{a(S) \mid S \subseteq Q\} \geq \sup\{a(S') \mid S' \subseteq Q'\}$

$= \sup'$

($\sup'$ must exist)